

Introduction

In the contemporary banking sector, the financial institutions aspire to raise profit levels by transforming liability customers (depositors) to asset customers (borrowers). One of these practices is the provision of personal loans to the appropriate customers.

This work will attempt to forecast the customers who are likely to take a personal loan based on historical data. The k-Nearest Neighbours (k-NN) classification model is applied in order to assist focused marketing tactics and enhance the efficacy of campaigns.

Dataset Description

The data set comprises of 5000 records of customers with demographic and financial information (age, income, family size, and credit card usage).

Personal Loan is the target variable and it denotes whether a customer accepted the loan offering. There was an imbalance of in the classes since only about 9.6% of customers took up the loan.

Data Preprocessing

These are some of the preprocessing steps that had been undertaken prior to model development.

To begin with, the irrelevant attributes, including ID and ZIP Code were eliminated, since they are not relevant in prediction.

The target variable Personal Loan was then cast into a categorical variable.

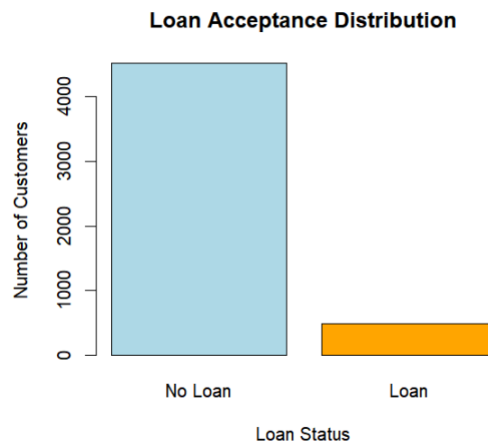
All numerical variables were then normalized to make the same contribution in distance calculations as necessary in the k-NN algorithm.

Lastly, the data set was separated into:

- 60% training data
- 40% validation data

Data Visualization & Insights

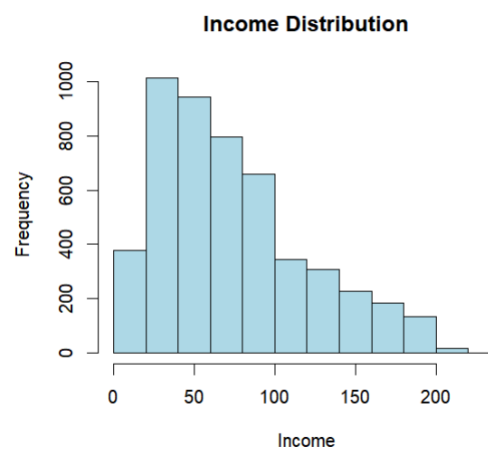
- **Bar Chart: Loan Acceptance Distribution**



Explanation :

The bar chart shows that the number of customers who did not accept the loan is significantly higher than those who did. This confirms that the dataset is imbalanced.

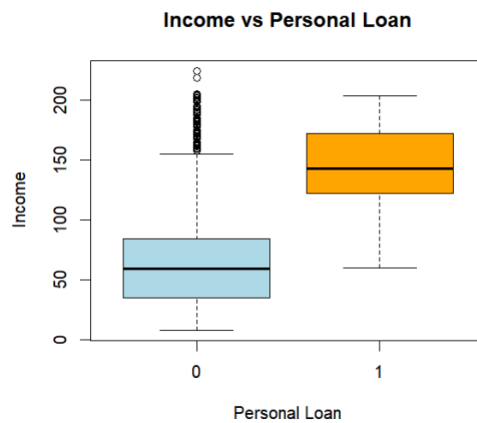
- **Histogram: Income Distribution**



Explanation:

The histogram shows the distribution of customer income. It can be observed that customers with higher income levels are more likely to accept personal loans.

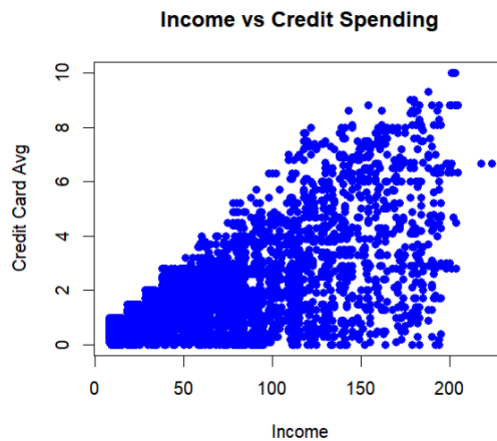
- **Boxplot: Income vs Personal Loan**



Explanation:

The boxplot indicates that customers who accepted the loan have a higher median income compared to those who did not. This suggests that income is a strong predictor of loan acceptance.

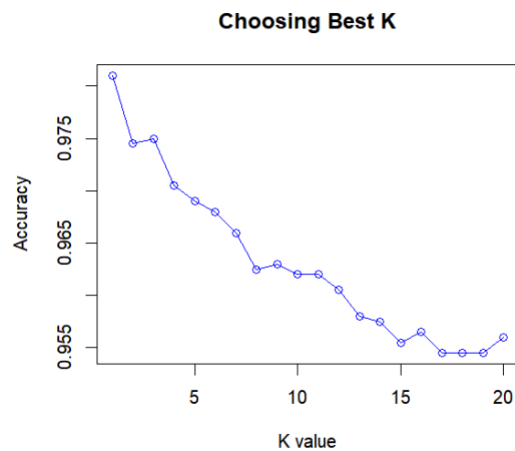
- **Scatterplot: Income vs CCAvg**



Explanation:

The scatterplot shows a positive relationship between income and credit card spending. Customers with higher spending behavior tend to be more likely to accept loans.

Choosing the Best k



The best value of k was identified by trying the values between 1 and 20 and using the accuracy at validating the data. The findings indicate that the largest accuracy was attained through $k = 1$ with accuracy of 0.981 (98.1%).

Even though $k = 1$ is the most accurate, it can result in overfitting since the model can be very sensitive to noise and outliers. Even with this shortcoming, $k = 1$ has been chosen because it has the highest predictive power of this data..

Classification Results

```
> table(best_pred, valid$Personal.Loan)

best_pred    0    1
0 1792   32
1    6  170
> library(class)
> library(kknn)
>
> mean(best_pred == valid$Personal.Loan)
[1] 0.981
```

The results of the classification indicate that the model had an accuracy rate of 98.1% and this implies a good predictive behaviour. The confusion matrix shows that majority of the customers were appropriately categorized or put in their respective categories. Nonetheless, the accuracy might be high due to class imbalance in the sample, in which case most of the customers had not taken the loan..

Distance Metrics Comparison

Two distance metrics were used:

- Euclidean distance (standard k-NN)
- Weighted distance (kkn)

```
> mean(pred2 == valid$Personal.Loan)
[1] 0.979
```

Euclidean distance method attained the accuracy of 98.1% and weighted distance method attained the accuracy of 97.9%.

The findings show that the Euclidean distance was a little better in the performance compared to the weighted distance. Even though weighted k-NN assigns more weight to the closer neighbours, it did not significantly enhance the performance of the model in this case. The accuracy difference between the two methods is not very large and this indicates that both distance measures are suitable in this dataset..

Model Evaluation (Cross-Validation)

```
k-Nearest Neighbors
3000 samples
6 predictor
2 classes: '0', '1'

No pre-processing
Resampling: Cross-Validated (10 fold)
Summary of sample sizes: 2700, 2700, 2700, 2699, 2701, 2700, ...
Resampling results across tuning parameters:

k   Accuracy   Kappa
5   0.9693377   0.7845552
7   0.9630044   0.7325779
9   0.9619988   0.7215722
11  0.9603344   0.7063555
13  0.9583333   0.6875996
15  0.9560010   0.6647285
17  0.9536677   0.6419948
19  0.9520010   0.6244256
21  0.9513344   0.6174030
23  0.9486677   0.5901600

Accuracy was used to select the optimal model using the largest value.
The final value used for the model was k = 5.
> View(model_cv)
> max(model_cv$results$Accuracy)
[1] 0.9693377
```

A cross-validation that includes 10 folds was conducted to determine the performance of the model. The findings show that the best value of k is 5 and cross-validation accuracy is 96.9%.

This implies that the model works uniformly on various subsets of the data, and is not too sensitive to a particular training set. The observation made is that the value of k that has been obtained through cross-validation (k = 5) is not the same as the one that has been obtained with the help of the validation set (k = 1). This means that k = 1 is more accurate but k = 5 is a more generalized model..

Discussion

Advantages:

- Simple and easy to implement
- No training phase required
- Works well with nonlinear data

Disadvantages:

- Sensitive to irrelevant features
- Requires normalization
- Slower for large datasets

Conclusion

This study successfully applied the k-Nearest Neighbours (k-NN) algorithm to predict customer acceptance of personal loans. The model achieved a high validation accuracy of 98.1%, with an optimal k value of 1.

However, cross-validation results suggested that $k = 5$ provides a more stable and generalized model with an accuracy of 96.9%.

Key factors influencing loan acceptance include income, credit card usage, and education level. Overall, the model demonstrates strong predictive capability and can assist banks in improving targeted marketing strategies.

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Muhd Aqil Irsyaad Bin Nor Azlisham	22010826	Data preprocessing
Muhammad Akmal Harith Bin Azlan	22010886	Model building
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Muhammad Iman Bin Shahrul Azizi	22011681	Report writing

Google Drive Link :

<https://drive.google.com/drive/folders/1TtSYSDHnX0scpoMLVnfWP9bvAVyKvOPh?usp=sharing>